Pergamon

Tetrahedron Letters 42 (2001) 4075–4077

TETRAHEDRON LETTERS

$CuBr_2$-mediated direct aqueous bromolactonization of 2,3-allenoates. An efficient access to β-bromobutenolides

Shengming Ma* and Shulin Wu

State Key Laboratory of Organometallic Chemistry, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 354 Fenglin Lu, Shanghai 200032, PR China

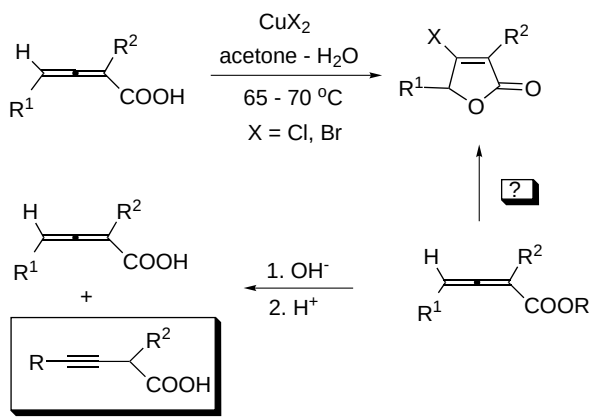
Received 23 November 2000; revised 5 April 2001; accepted 12 April 2001

Abstract—An efficient procedure for the synthesis of β-bromobutenolides was developed. 2,3-Allenoates with different substitution patterns react with $CuBr_2$ in aqueous ethanol at 80–85°C to afford the corresponding β-bromobutenolides in moderate to excellent yields (up to 97%). © 2001 Elsevier Science Ltd. All rights reserved.

Butenolides can be found in many natural products with an unusual range of biological activities.[1] Since butenolide-containing compounds have been considered as potential insecticides, bactericides, antibiotics, anticancer agents, anti-inflammatories, allergy inhibitors, antisoriasis agents, cyclooxygenase inhibitors and phospholipase A_2 inhibitors, etc.[2] this unit also represents a structural feature of compounds of pharmaceutical interest. In addition, they appear to be versatile intermediates[3,4] for the synthesis of cyclic and acyclic systems. Recently, much attention has been focused on the efficient and diverse synthesis of these valuable compounds,[4] particularly β-halobutenolides.[5] This is, in large part, due to biologically active natural products[6] isolated from marine sources which have β-halobutenolides as their structural moieties.

During the course of our project aimed at exploration of new synthetic uses of functionalized allenes,[7] we have recently developed a high-yielding procedure for the synthesis of β-halobutenolides from 2,3-allenoic acids and CuX_2 (X = Br or Cl) (Scheme 1).[8] In this transformation, most of the starting materials, i.e. the 2,3-allenoic acids, were obtained from the corresponding esters by hydrolysis. In some cases the yields were low.[9] Furthermore, the concomitant formation of 3-alkynoic acids was often observed, even as the sole products in some cases (Scheme 1).[10] Hence, it was highly desirable to develop new methodologies using the esters of 2,3-allenoic acids directly as the starting materials.

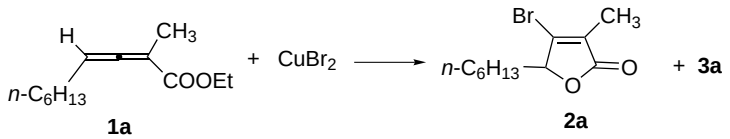
2,3-Allenoates with different substitution patterns are easily available through Wittig-type reactions.[11] As a starting point, the reaction conditions ($CuBr_2$, in a 2:1 mixture of acetone/H_2O, 65–70°C) for the halolactonization of 2,3-allenoic acids established previously[8] were used to cyclize ethyl 2-methyl-2,3-decadienoate (**1a**) to afford the corresponding β-bromobutenolide **2a** in a 62% yield within 12 h, together with recovery of **1a** in a 27% yield. After screening, we found that the reaction carried out in an ethanol/water (3:2) medium at 80–85°C gave better results. The starting material **1a** was consumed completely and β-bromobutenolide **2a** was isolated in an 86% yield (Scheme 2). It is noteworthy that bromolactonization of **1a** with NBS under the conditions described in the literature gave the product **2a** only in 53% yield together with 15% of 3-bromo-4-oxo-2-decenoate.[3a,12]



Scheme 1.

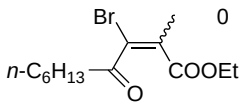
* Corresponding author.

0040-4039/01/$ - see front matter © 2001 Elsevier Science Ltd. All rights reserved.
PII: S0040-4039(01)00634-7



1a 2a

conditions	recovery of **1a**	yield of **2a** +	yield of **3a**
65-70 °C/acetone/H_2O/12 h	27%	62%	0
80-85 °C/ethanol/H_2O/12 h	0	86%	0
NBS/H_2O/rt/14 h	0	53%	15%



3a

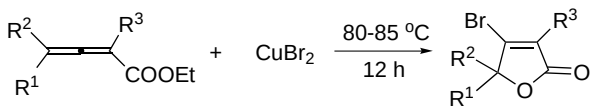
Scheme 2.

With these conditions in hand, this halolactonization reaction was examined for differently substituted 2,3-allenoates, and the results are summarized in Table 1. In all cases, the starting allenoates **1** reacted completely, and the reaction proceeded smoothly and cleanly to give the corresponding β-bromobutenolides **2** in moderate to excellent yields, the highest being 97%. Generally, the yields of 3,5-dialkyl bromobutenolides are higher than those of 3,5,5-trialkyl bromobutenolides (compare entries 1–7 with 8–9). On the other hand, it was previously reported that the bromolactonization of ethyl 2,4-dimethyl-2,3-pentadienoate **1i** with molecular Br_2 afforded the corresponding bromobutenolides **2i** in only a 58% yield.[13] Of special interest to us was that the allenic ester **1k**, which bears an allyl group at the 2-position, reacted with $CuBr_2$ also to give the β-bromobutenolide **2k** in a moderate yield (entry 10). Dibromination of the allylic C=C bond[14] was not observed. Products like **2k**, having different types of functional groups at the 3- and 4-positions should provide efficient access to other butenolide derivatives via further elaboration.

In an attempt to bromolactonize ethyl 4-aryl-2,3-butedienoates under these reaction conditions, rather disappointing results were obtained. For example, from ethyl 2-methyl-4-phenyl-2,3-butedienoate **1l**, the reaction afforded **2l** in 40% yield together with an unidentified product. Nevertheless, when the benzyl ester **1l′** was used instead of the ethyl ester **1l**, the yield of **2l** was improved to 67%. Similarly, bromolactonization of benzyl 2-methyl-4-phenyl-2,3-hexadienoate **1m′** afforded **2m** in 66% yield (Scheme 3).

In conclusion, we have developed a convenient and efficient method for the synthesis of β-bromobutenolides, i.e. the reaction of easily available 2,3-allenoates with $CuBr_2$ in aqueous ethanol. This reaction opens up

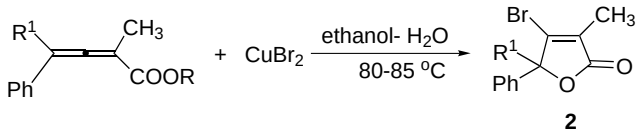
Table 1. Bromolactonization of allenoates with $CuBr_2$[a]



1 2

Entry	**1**			Product	Yield (%)
	R^1	R^2	R^3		
1	n-C_6H_{13}	H	n-C_3H_7 (**1b**)	**2b**	95
2	n-C_6H_{13}	H	$C_6H_5CH_2$ (**1c**)	**2c**	97
3	n-C_3H_7	H	$C_6H_5CH_2$ (**1d**)	**2d**	78
4	n-C_3H_7	H	n-C_3H_7 (**1e**)	**2e**	68
5	n-C_3H_7	H	CH_3 (**1f**)	**2f**	86
6	CH_3	H	CH_3 (**1g**)	**2g**	72
7	CH_3	H	n-C_3H_7 (**1h**)	**2h**	84
8	CH_3	CH_3	CH_3 (**1i**)	**2i**	68
9	CH_3	CH_3	n-C_3H_7 (**1j**)	**2j**	66
10	n-C_3H_7	H	Allyl (**1k**)	**2k**	54

[a] The reaction was carried out using **1** (0.5 mmol), $CuBr_2$ (4 equiv.), and ethanol:H_2O (4 mL, 3:2).



2

1l (R^1 = H, R = C_2H_5) **2l** 40% + unidentified product

1l' (R^1 = H, R = $CH_2C_6H_5$) **2l** 67%

1m' (R^1 = C_2H_5, R = $CH_2C_6H_5$) **2m** 66%

Scheme 3.

new possibilities for the synthesis of optically active butenolides due to the easy availability of optically active 2,3-allenoates.[10] Further studies on the scope and synthetic application of this methodology are being carried out in our laboratory.

Acknowledgements

We thank the Major State Basic Research Development Program (Grant No. G 2000077500), the National Natural Science Foundation of China (Grant No. 29932020), and the Postdoctoral Foundation of Shanghai for Financial Support. Shengming Ma is the recipient of the 1999 Qiu Shi Award for Young Scientific Workers issued by the Hong Kong Qiu Shi Foundation of Science and Technology (1999–2002) and the Special Grant for Outstanding Young Scientists issued by Shanghai Municipal Committee of Science and Technology (OOXD 14028).

References

1. Larock, R. D.; Riefling, B.; Fellows, C. A. *J. Org. Chem.* **1978**, *43*, 131 and references cited therein. For recent examples, see: (a) Chia, Y.; Chang, F.; Wu, Y. *Tetrahedron Lett.* **1999**, *40*, 7513; (b) Takahashi, S.; Maeda, K.; Hirota, S.; Nakata, T. *Org. Lett.* **1999**, *1*, 2025; (c) Casiraghi, G.; Zanardi, F.; Battistini, L.; Rassu, G.; Appendino, G. *Chemtracts* **1998**, *11*, 803; (d) Siddiqui, B. S.; Afshan, F.; Faizi, S.; Naqvi, S. N.-H.; Tariq, R. M. *J. Chem. Soc., Perkin Trans. 1* **1999**, 2367; (e) Ostuka, H.; Kotani, K.; Bando, M.; Kido, M.; Takeda, Y. *Chem. Pharm. Bull.* **1998**, *46*, 1180; (f) Ishikawa, T.; Nishigaya, K.; Uchikoshi, H.; Chen, I. *J. Nat. Prod.* **1998**, *64*, 534; (g) Driol, S.; Felluga, F.; Forzeto, C.; Nitti, P.; Pitacco, G.; Valentin, E. *J. Org. Chem.* **1998**, *63*, 2385.
2. Ma, S.; Shi, Z. *J. Org. Chem.* **1998**, *63*, 6387 and references cited therein.
3. (a) Hanessian, S. In *The Total Synthesis of Natural Products: The Chiron Approach*; Pergamon Press: New York, 1983; Chapter 9; (b) Stork, G.; Rychnovsky, S. D. *J. Am. Chem. Soc.* **1987**, *109*, 1564; (c) Dankwardt, S. M.; Dankwardt, J. W.; Schlessinger, R. H. *Tetrahedron Lett.* **1998**, *39*, 4971, 4975 and 4979.
4. For a review, see: Rao, Y. S. *Chem. Rev.* **1976**, *76*, 625. Knight, D. W. *Contemp. Org. Synth.* **1994**, 287.
5. (a) Shingu, K.; Hagishita, S.; Nakagawa, M. *Tetrahedron Lett.* **1967**, 4731; (b) Kresze, G.; Kloimsteion, L.; Runge, W. *Liebigs Ann. Chem.* **1976**, 979; (c) Bravermman, S.; Reisman, D. *Tetrahedron Lett.* **1977**, 1753; (d) Ma, S.; Shi, Z.; Yu, Z. *Tetrahedron Lett.* **1999**, *40*, 2393; (e) Ma, S.; Shi, Z.; Yu, Z. *Tetrahedron* **1999**, *55*, 12137.
6. (a) Kazlauskas, K.; Murphy, P. T.; Qinn, R. J.; Wells, R. J. *Tetrahedron Lett.* **1977**, 37; (b) Pettus, J. A.; Wing, R. M.; Sims, J. J. *Tetrahedron Lett.* **1977**, 41; (c) Ohta, K. *Agric. Biol. Chem.* **1977**, *41*, 2105; (d) McCombs, J. D.; Blunt, J. W.; Chambers, M. V.; Munro, M. H.; Robinson, W. T. *Tetrahedron* **1988**, *44*, 1489; (e) de Nys, R.; Coll, J. C.; Bowden, B. F. *Aust. J. Chem.* **1992**, *45*, 1625; (f) de Nys, R.; Wright, A. D.; König, G. M.; Sticher, O. *Tetrahedron* **1993**, *49*, 11213.
7. (a) Ma, S.; Zhao, S. *J. Am. Chem. Soc.* **1999**, *121*, 7943; (b) Ma, S.; Zhao, S. *Org. Lett.* **2000**, *2*, 2495; (c) Ma, S.; Zhang, J. *Chem. Commun.* **2000**, 117; (d) Ma, S.; Li, L. *Org. Lett.* **2000**, *2*, 941; (e) Ma, S.; Shi, Z.; Yu, Z. *Tetrahedron Lett.* **1999**, *40*, 2393; (f) Ma, S.; Shi, Z.; Yu, Z. *Tetrahedron* **1999**, *55*, 12137; (g) Ma, S.; Xie, H. *Org. Lett.* **2000**, *2*, 3801; (h) Ma, S.; Duan, D.; Shi, Z. *Org. Lett.* **2000**, *2*, 1419.
8. Ma, S.; Wu, S. *J. Org. Chem.* **1999**, *64*, 9314.
9. (a) Bestman, H.-J.; Hartung, H. *Chem. Ber.* **1966**, *99*, 1198; (b) Kresze, G.; Runge, W.; Ruch, E. *Liebigs Ann. Chem.* **1972**, *756*, 112; (c) Runge, W.; Kresze, G.; Ruch, E. *Liebigs Ann. Chem.* **1975**, 1361.
10. Marshall, A.; Wolf, M. A.; Wallace, E. M. *J. Org. Chem.* **1997**, *62*, 367. In our case, hydrolysis of ethyl 2-methyl-2,3-decadienoate **1a** with NaOH led to the formation of a mixture of 2-methyl-2,3-decadienoic acid and 2-methyl-3-decynoic acid which cannot be separated thoroughly by chromatography on silica gel.
11. (a) Lang, R. W.; Hansen, H. J. *Helv. Chim. Acta* **1980**, *63*, 435; (b) Lang, R. W.; Hansen, H. J. *Org. Synth.* **1984**, *62*, 202.
12. (a) Font, J.; Gracia, A.; de March, P. *Tetrahedron Lett.* **1990**, *38*, 5517; (b) de March, P.; Font, J.; Gracia, A.; Zheng, Q. *J. Org. Chem.* **1995**, *60*, 1814.
13. Gill, G. B.; Idris, M. S. H. *Tetrahedron Lett.* **1985**, *26*, 4811.
14. (a) Uemura, S.; Okazaki, H.; Okano, M. *J. Chem. Soc., Perkin Trans. 1* **1978**, 1278; (b) Rodebaugh, R.; Debenham, J. S.; Freiser-Reid, B.; Synder, J. P. *J. Org. Chem.* **1999**, *64*, 1758.